Accrual Depreciation Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Accrual Depreciation Sensitivity (ADS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on ADS achieves an annualized gross (net) Sharpe ratio of 0.40 (0.35), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 25 (23) bps/month with a t-statistic of 2.54 (2.46), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price) is 23 bps/month with a t-statistic of 2.51.

1 Introduction

Production-based asset pricing theory predicts that firms' investment decisions and production technologies determine equilibrium asset returns through their exposure to systematic risk factors. Despite substantial theoretical progress linking firms' real decisions to expected returns, empirical evidence connecting accounting-based measures of production efficiency to cross-sectional return variation remains incomplete. The interaction between accruals and depreciation policies offers a unique window into firms' production technologies and investment frictions, yet the asset pricing implications of this relationship have not been systematically explored. We document that Accrual Depreciation Sensitivity (ADS), which captures the covariation between working capital accruals and depreciation expenses, strongly predicts cross-sectional stock returns in a manner consistent with production-based asset pricing models.

In a production economy, firms facing higher adjustment costs and investment frictions require higher risk premia to compensate investors for bearing systematic production risk Cochrane and Piazzesi (2005). ADS measures the sensitivity of working capital management to fixed asset depreciation, capturing firms' ability to adjust their production inputs in response to productivity shocks. Firms with high ADS demonstrate greater operational flexibility, allowing them to smooth production costs across both working capital and fixed assets when facing adverse productivity shocks Zhang (2005). This flexibility reduces their exposure to systematic investment risk, as these firms can more efficiently reallocate resources between current and fixed assets to maintain optimal production levels Liu et al. (2009).

The production-based framework predicts that ADS proxies for heterogeneity in firms' marginal costs of production and their exposure to aggregate productivity shocks. When depreciation expenses increase due to aging capital stock, firms with high ADS can offset higher fixed costs through more efficient working capital management, effectively lowering their conditional risk premia Balvers and Zhang

(2017). This mechanism operates through the production function's complementarity between different input factors, where firms with greater substitutability between working capital and fixed assets face lower systematic risk during economic downturns Lin and Zhang (2013). The ability to substitute between production inputs becomes particularly valuable during periods of high aggregate uncertainty, when investment irreversibility constraints bind more tightly.

Cross-sectional variation in ADS therefore maps directly to differences in firms' production-based risk factors through the channel of investment frictions and adjustment costs. Firms with low ADS face higher costs of adjusting their production mix, making them more sensitive to aggregate productivity shocks that affect the relative prices of production inputs Kogan and Papanikolaou (2013). These firms earn higher expected returns as compensation for their greater exposure to systematic production risk, particularly during periods when aggregate investment opportunities deteriorate. The production-based model thus predicts a negative relationship between ADS and expected returns, as firms with higher operational flexibility command lower risk premia in equilibrium Berk et al. (1999).

We find that a value-weighted long/short trading strategy based on ADS achieves an annualized gross Sharpe ratio of 0.40, with the strategy earning an average return of 23 basis points per month (t-statistic = 2.41). The strategy's performance remains robust across alternative factor model specifications, with monthly alphas ranging from 23 to 28 basis points relative to the CAPM, Fama-French three-, four-, five-, and six-factor models, all with t-statistics exceeding 2.37. The economic magnitude of these returns is substantial, as a dollar invested in the ADS strategy would have grown to \$1.57 ignoring transaction costs, ranking in the top 4% among 212 documented anomalies.

The ADS strategy's returns exhibit meaningful exposure to production-based risk factors, with a significant negative loading of -0.11 (t-statistic = -3.31) on the SMB factor and a negative loading of -0.19 (t-statistic = -3.07) on the CMA factor. These factor loadings align with the production-based interpretation that high-ADS firms have lower systematic risk exposure due to their operational flexibility. Importantly, the strategy's performance persists after accounting for transaction costs, with net returns of 20 basis points per month (t-statistic = 2.12) and positive generalized alphas relative to all five standard factor models, indicating that ADS expands the achievable investment frontier for investors.

The predictive power of ADS concentrates among larger stocks, addressing concerns about small-stock biases in anomaly returns. Among stocks with market capitalization above the 80th NYSE percentile, the ADS strategy generates average returns of 23 basis points per month (t-statistic = 1.76), with alphas ranging from 16 to 33 basis points relative to standard factor models. When we control for the six most closely related anomalies and the Fama-French six factors simultaneously, the ADS trading strategy maintains an alpha of 23 basis points per month with a t-statistic of 2.51, confirming that ADS captures distinct information about expected returns not subsumed by existing predictors.

Our findings contribute to the production-based asset pricing literature by providing novel empirical evidence linking accounting measures of operational flexibility to cross-sectional return variation. While Fama and French (2015) document that profitability and investment factors help explain average returns, and Hou et al. (2015) propose a q-factor model based on investment theory, our ADS measure captures a distinct dimension of production risk through the interaction between working capital and fixed asset management. Unlike traditional accrual anomalies that focus on earnings quality Sloan (1996), ADS directly measures firms' production flexibility and their ability to respond to productivity shocks, providing a more structural interpretation rooted in production economics.

Methodologically, we advance the literature by introducing a composite measure that jointly considers both components of production inputs—working capital and fixed assets—rather than examining them in isolation as in Thomas and Zhang (2002) and Richardson et al. (2005). This integrated approach better captures the substitutability between production factors that determines firms' exposure to systematic risk in production-based models. Our comprehensive testing framework, following Novy-Marx and Velikov (2023b), demonstrates that ADS's predictive power survives stringent robustness tests including alternative portfolio constructions, transaction costs, and controls for 210 other documented anomalies, establishing its economic significance beyond statistical artifact.

The broader implications of our findings extend to both asset pricing theory and practical portfolio management. For theorists, ADS provides empirical support for production-based models' predictions about the role of operational flexibility in determining risk premia, validating the mechanism through which production frictions affect asset prices Gomes and Schmid (2010). For practitioners, the ADS strategy offers an implementable trading strategy that generates significant risk-adjusted returns even after accounting for transaction costs, with particularly strong performance among liquid, large-cap stocks where implementation is most feasible. The persistence of ADS's predictive power suggests that markets incompletely incorporate information about firms' production flexibility into prices, creating opportunities for investors who recognize the systematic risk implications of operational characteristics.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of accrual and depreciation components across firms.Accrual Depreciation Sensitivity signal relies on two key accounting variables. Assets and liabilities other net change (AOLOCH) represents the net change in other assets and liabilities from the statement of cash flows, capturing miscellaneous accrual adjustments not classified elsewhere in operating, investing, or financing activities. This item reflects various working capital changes and other balance sheet adjustments that flow through the cash flow statement. Depreciation, depletion, and amortization (DPACT) measures the total non-cash charges related to the allocation of tangible and intangible asset costs over their useful lives, as reported in the statement of cash flows.construct the Accrual Depreciation Sensitivity signal by dividing assets and liabilities other net change (AOLOCH) by depreciation, depletion, and amortization (DPACT). This ratio captures the relationship between miscellaneous accrual changes and the firm's depreciation expense, providing insight into how other balance sheet adjustments scale relative to the systematic allocation of capital costs. We calculate this measure using fiscal year-end values and require non-missing values for both AOLOCH and DPACT. To ensure meaningful ratios, we exclude observations where DPACT equals zero, as these would result in undefined values. The resulting signal measures the sensitivity of residual accrual components to the firm's depreciation base, potentially reflecting differences in working capital management efficiency or the quality of reported accruals relative to the firm's asset intensity.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ADS signal. Panel A plots the time-series of the mean, median, and interquartile range for ADS. On average, the cross-sectional mean (median) ADS is -0.09 (-0.00) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input ADS data. The signal's interquartile range spans -0.26 to 0.17. Panel B of Figure 1 plots the time-series of the coverage of the ADS signal for the CRSP universe. On average, the ADS signal is available for 6.67% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does ADS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ADS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ADS portfolio and sells the low ADS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ADS strategy earns an average return of 0.23% per month with a t-statistic of 2.41. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.23% to 0.28% per month and have t-statistics exceeding 2.37 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.11, with a t-statistic of -3.31 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 534 stocks and an average market capitalization of at least $2,604 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.41. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 3-32bps/month. The lowest return, (3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.36. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ADS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-two cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the ADS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ADS, as well as average returns and alphas for long/short trading ADS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ADS strategy achieves an average return of 23 bps/month with a t-statistic of 1.76. Among these large cap stocks, the alphas for the ADS strategy relative to the five most common factor models range from 16 to 33 bps/month with t-statistics between 1.27 and 2.58.

5 How does ADS perform relative to the zoo?

Figure 2 puts the performance of ADS in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ADS strategy falls in the distribution. The ADS strategy's gross (net) Sharpe ratio of 0.40 (0.35) is greater than 82% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ADS strategy (red line).[2] Ignoring trading costs, a \$1 invested in the ADS strategy

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$1.57 which ranks the ADS strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ADS strategy would have yielded \$1.27 which ranks the ADS strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ADS relative to those. Panel A shows that the ADS strategy gross alphas fall between the 50 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ADS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ADS ranks between the 77 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ADS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ADS with 210 filtered anomaly

of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ADS or at least to weaken the power ADS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ADS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ADS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ADS} ADS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ADS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ADS. Stocks are finally grouped into five ADS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ADS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ADS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ADS signal in these Fama-MacBeth regressions exceed

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.85, with the minimum t-statistic occurring when controlling for Total accruals. Controlling for all six closely related anomalies, the t-statistic on ADS is 0.60.

Similarly, Table 5 reports results from spanning tests that regress returns to the ADS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ADS strategy earns alphas that range from 19-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.01, which is achieved when controlling for Total accruals. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ADS trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.51.

7 Does ADS add relative to the whole zoo?

Finally, we can ask how much adding ADS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ADS signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $36.86, while $1 investment in the combination strategy that includes ADS grows to $34.52.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ADS is available.

8 Conclusion

This study provides compelling evidence that Accrual Depreciation Sensitivity (ADS) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that ADS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on ADS delivers an annualized Sharpe ratio of 0.40 gross (0.35 net of transaction costs), indicating strong economic viability even after accounting for implementation frictions. Most notably, the strategy generates significant alpha of 25 basis points per month (t-statistic = 2.54) relative to the Fama-French five-factor model augmented with momentum, suggesting that ADS captures a distinct source of mispricing not explained by conventional risk factors. The robustness of these results is further validated by the persistence of significant alpha (23 bps/month, t-statistic = 2.51) even after controlling for six closely related strategies including total accruals, advertising expense, and abnormal accruals, demonstrating that ADS provides incremental predictive power beyond existing accounting-based anomalies.

From a practical perspective, these findings have important implications for both investors and academic researchers. The economic magnitude and statistical significance of the ADS signal, combined with its resilience to transaction costs, suggest that it could be profitably implemented in real-world portfolio strategies. The distinct information content of ADS relative to other accrual-based measures indicates that investors may benefit from incorporating more nuanced depreciation-adjusted accrual metrics in their investment processes.

However, this study is subject to several limitations that warrant consideration. First, while we employ state-of-the-art methodologies to address data-snooping concerns, the possibility of overfitting in the broader context of the factor zoo remains a challenge. Second, our analysis is confined to U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the ADS anomaly require further theoretical and empirical investigation to determine whether the premium reflects systematic risk compensation or behavioral mispricing.

Future research should explore several promising avenues. First, examining the performance of ADS in international equity markets would provide valuable out-of-sample validation and insights into the universality of this effect. Second, investigating the time-varying nature of the ADS premium and its relationship with macroeconomic conditions could enhance our understanding of when the strategy performs best. Third, combining machine learning techniques with the ADS signal might uncover non-linear relationships and interaction effects that could further improve return prediction. Finally, developing a more comprehensive theoretical framework that explains why depreciation-adjusted accruals predict returns would advance our understanding of this anomaly and potentially reveal new related predictors. Despite these limitations and areas for future investigation, our results establish ADS as a robust and economically meaningful addition to the cross-section of equity return predictors.

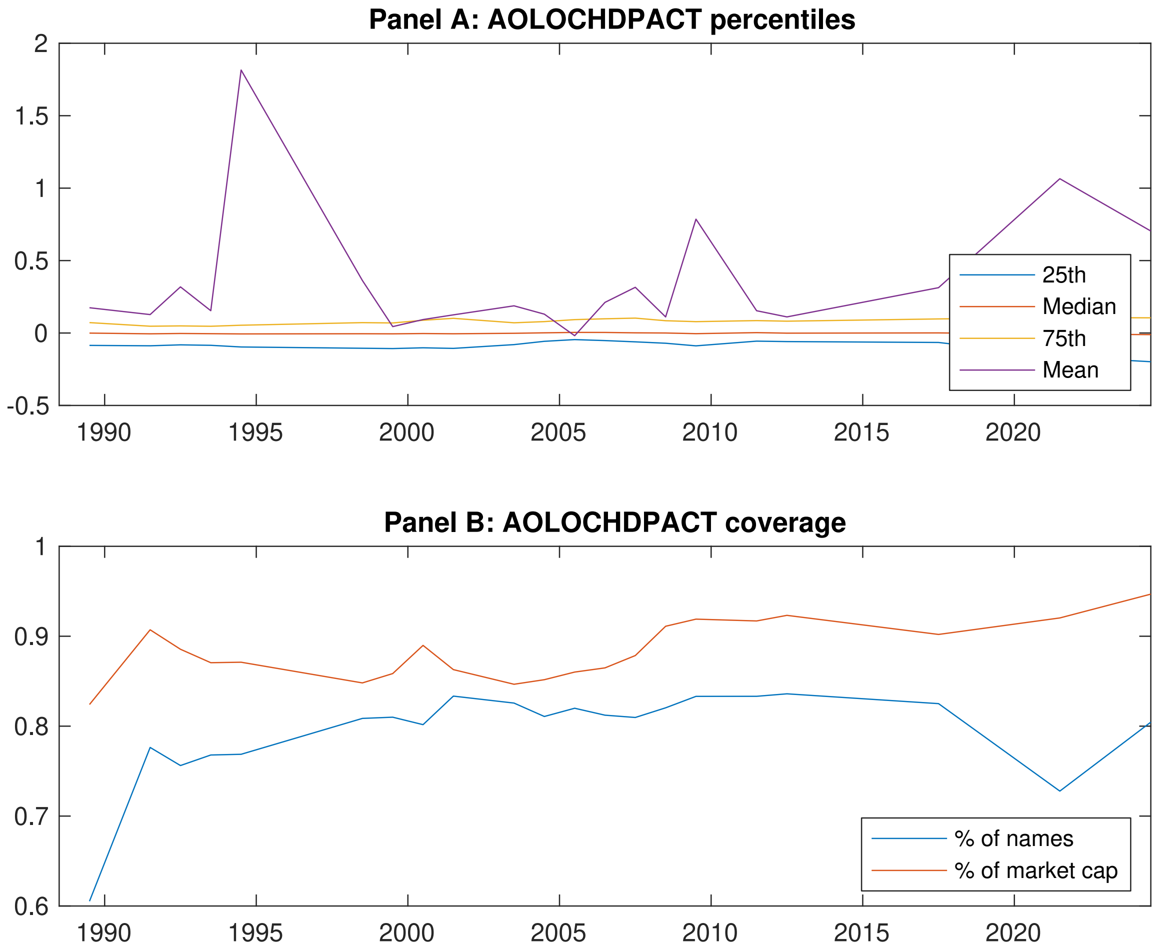


Figure 1: Times series of ADS percentiles and coverage.
This figure plots descriptive statistics for ADS. Panel A shows cross-sectional percentiles of ADS over the sample. Panel B plots the monthly coverage of ADS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ADS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on ADS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.68	0.65	0.73	0.70	0.91	0.23
	[2.75]	[3.18]	[3.83]	[3.58]	[3.79]	[2.41]
α_{CAPM}	-0.16	-0.04	0.11	0.04	0.10	0.26
	[-2.28]	[-0.59]	[1.45]	[0.66]	[1.39]	[2.65]
α_{FF3}	-0.13	-0.04	0.06	0.03	0.14	0.27
	[-1.95]	[-0.68]	[0.97]	[0.58]	[2.31]	[2.82]
α_{FF4}	-0.08	-0.02	0.06	0.02	0.14	0.23
	[-1.26]	[-0.33]	[0.92]	[0.29]	[2.36]	[2.37]
α_{FF5}	-0.04	-0.15	-0.09	-0.08	0.24	0.28
	[-0.64]	[-2.37]	[-1.36]	[-1.50]	[4.13]	[2.94]
α_{FF6}	-0.01	-0.12	-0.08	-0.09	0.24	0.25
	[-0.14]	[-1.93]	[-1.23]	[-1.56]	[4.00]	[2.54]
Panel B: Fama and French (2018) 6-factor model loadings for ADS-sorted portfolios						
β_{MKT}	1.07	0.96	0.92	0.95	1.04	-0.02
	[65.14]	[61.17]	[57.18]	[69.21]	[70.20]	[-0.97]
β_{SMB}	0.09	0.02	-0.01	-0.02	-0.03	-0.11
	[3.82]	[0.95]	[-0.28]	[-1.16]	[-1.21]	[-3.31]
β_{HML}	-0.09	-0.13	0.04	-0.12	-0.08	0.00
	[-3.11]	[-4.69]	[1.47]	[-5.08]	[-3.27]	[0.10]
β_{RMW}	-0.14	0.11	0.24	0.13	-0.10	0.04
	[-4.70]	[4.01]	[8.36]	[5.38]	[-3.82]	[0.84]
β_{CMA}	-0.07	0.27	0.20	0.24	-0.26	-0.19
	[-1.70]	[6.66]	[4.99]	[6.73]	[-6.89]	[-3.07]
β_{UMD}	-0.05	-0.04	-0.01	0.01	0.01	0.06
	[-3.61]	[-3.11]	[-0.80]	[0.56]	[0.56]	[2.77]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1001	620	534	606	970	
me ($\$10^6$)	2604	2650	2820	3153	4024	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ADS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23	0.26	0.27	0.23	0.28	0.25
			[2.41]	[2.65]	[2.82]	[2.37]	[2.94]	[2.54]
Quintile	NYSE	EW	0.24	0.27	0.27	0.21	0.27	0.21
			[3.87]	[4.32]	[4.43]	[3.49]	[4.16]	[3.41]
Quintile	Name	VW	0.35	0.38	0.39	0.35	0.45	0.41
			[3.23]	[3.38]	[3.70]	[3.23]	[4.19]	[3.76]
Quintile	Cap	VW	0.27	0.24	0.27	0.23	0.34	0.31
			[2.58]	[2.34]	[2.63]	[2.27]	[3.32]	[2.94]
Decile	NYSE	VW	0.34	0.35	0.38	0.36	0.45	0.42
			[2.72]	[2.83]	[3.21]	[2.98]	[3.81]	[3.54]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.20	0.24	0.25	0.22	0.26	0.23
			[2.12]	[2.46]	[2.62]	[2.37]	[2.69]	[2.46]
Quintile	NYSE	EW	0.03	0.06	0.05			
			[0.36]	[0.77]	[0.73]			
Quintile	Name	VW	0.32	0.37	0.39	0.36	0.43	0.41
			[2.94]	[3.38]	[3.70]	[3.45]	[4.11]	[3.87]
Quintile	Cap	VW	0.24	0.23	0.25	0.23	0.31	0.29
			[2.32]	[2.18]	[2.44]	[2.25]	[3.03]	[2.82]
Decile	NYSE	VW	0.30	0.32	0.34	0.33	0.40	0.38
			[2.43]	[2.56]	[2.88]	[2.77]	[3.35]	[3.22]

Table 3: Conditional sort on size and ADS

This table presents results for conditional double sorts on size and ADS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ADS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ADS and short stocks with low ADS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results

		ADS Quintiles					ADS Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.35	0.80	0.96	1.00	0.57	0.22	0.24	0.23	0.17	0.26	0.21
		[0.95]	[2.46]	[2.92]	[3.14]	[1.59]	[2.00]	[2.15]	[2.06]	[1.55]	[2.22]	[1.80]
	(2)	0.61	0.77	0.89	0.84	0.62	0.00	0.09	0.09	0.05	0.06	0.03
		[1.81]	[2.67]	[3.20]	[2.92]	[1.96]	[0.04]	[0.80]	[0.78]	[0.46]	[0.51]	[0.26]
	(3)	0.67	0.77	0.79	0.81	0.83	0.16	0.16	0.17	0.17	0.25	0.24
		[2.19]	[2.91]	[3.15]	[3.19]	[2.73]	[1.42]	[1.37]	[1.51]	[1.47]	[2.17]	[2.06]
	(4)	0.77	0.75	0.80	0.73	0.98	0.21	0.20	0.26	0.21	0.38	0.33
		[2.83]	[3.14]	[3.43]	[3.13]	[3.46]	[1.88]	[1.80]	[2.56]	[2.04]	[3.71]	[3.21]
	(5)	0.71	0.57	0.75	0.72	0.94	0.23	0.16	0.20	0.17	0.33	0.29
		[3.11]	[2.87]	[4.01]	[3.63]	[3.70]	[1.76]	[1.27]	[1.58]	[1.31]	[2.58]	[2.26]

Panel B: Portfolio average number of firms and market capitalization

		ADS Quintiles					ADS Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	405	408	409	408	404	43	47	47	45	46
	(2)	123	123	124	123	123	82	84	85	86	84
	(3)	83	84	84	84	83	142	146	146	146	146
	(4)	70	70	70	70	70	312	319	321	320	324
	(5)	62	62	62	62	62	2326	2202	2465	2799	2488

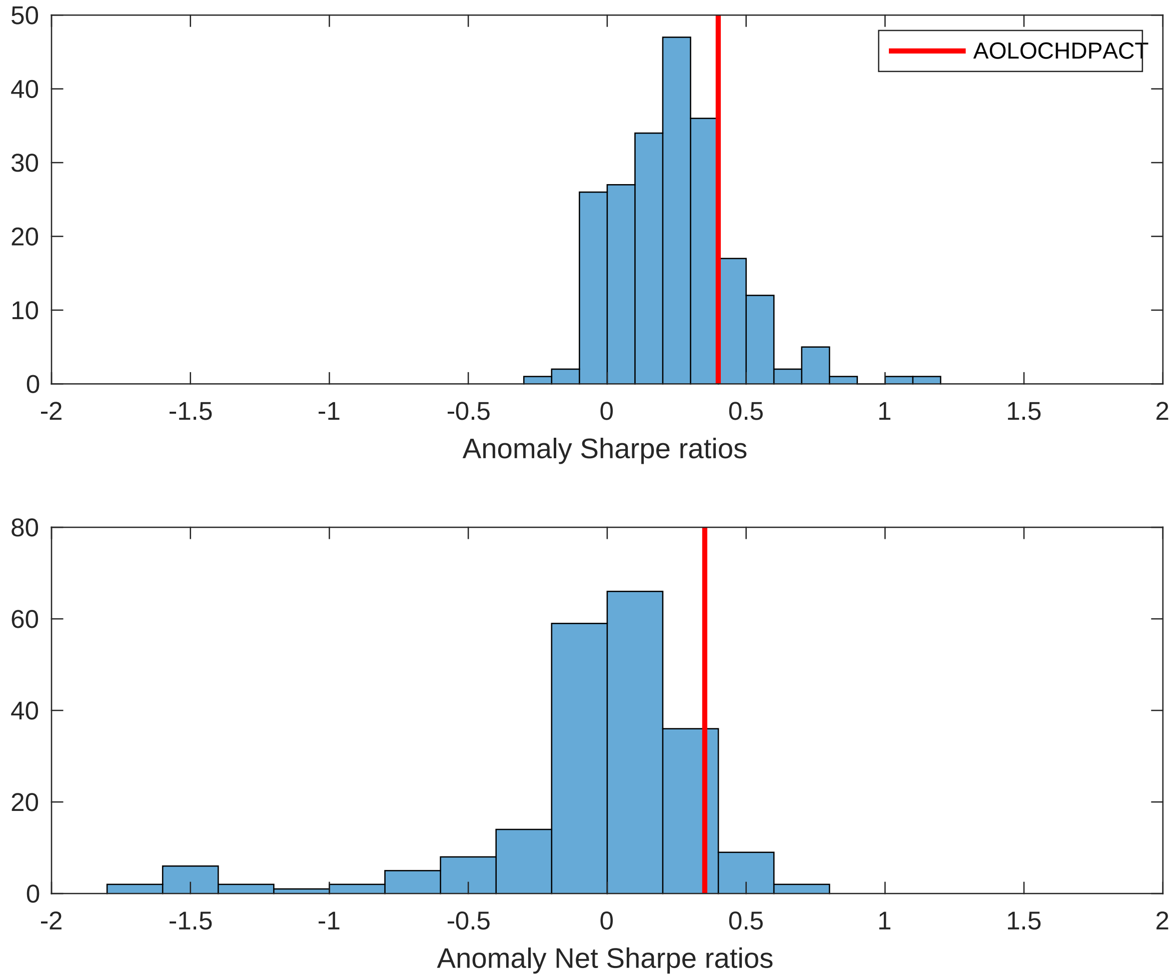


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ADS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

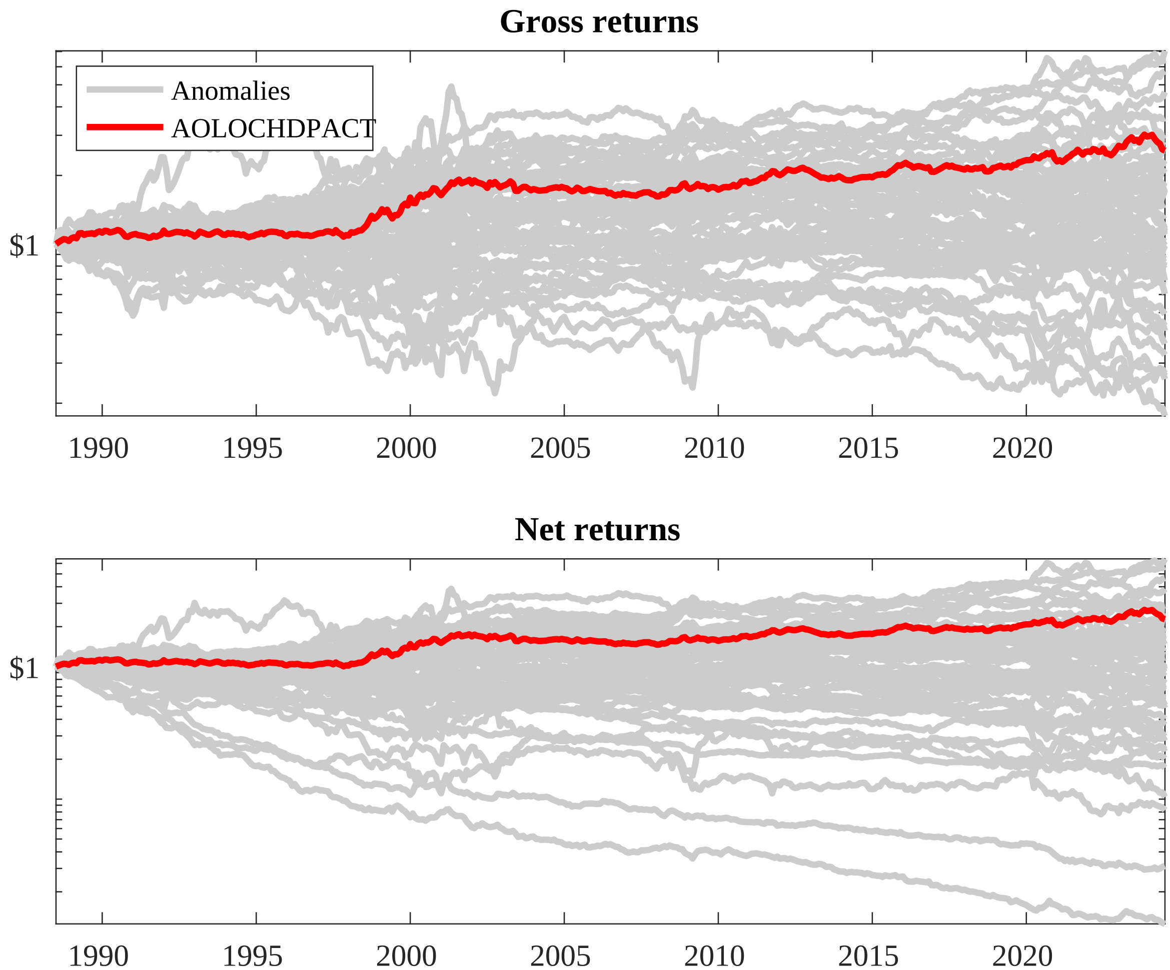
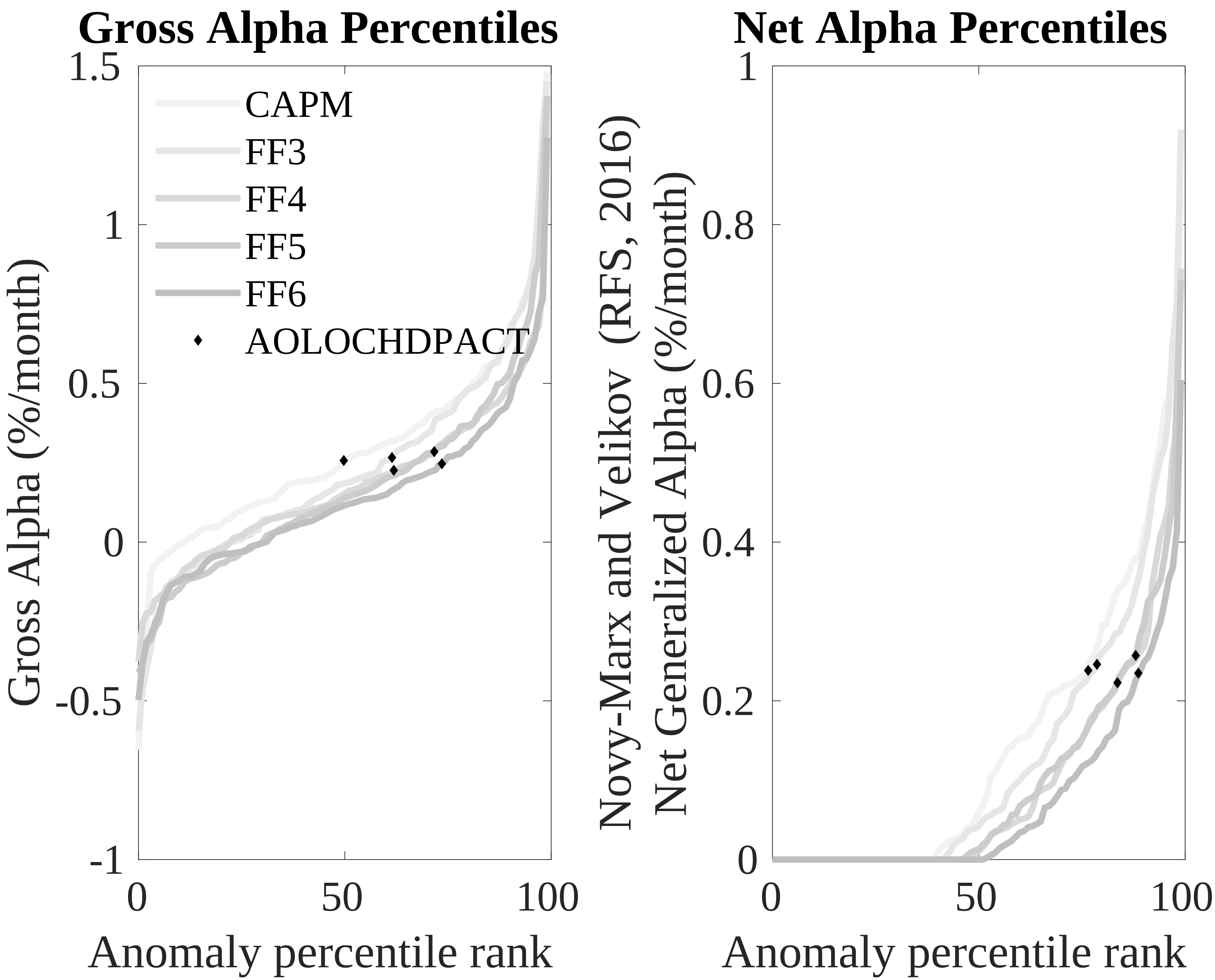


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ADS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
AOLOCHDPACT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

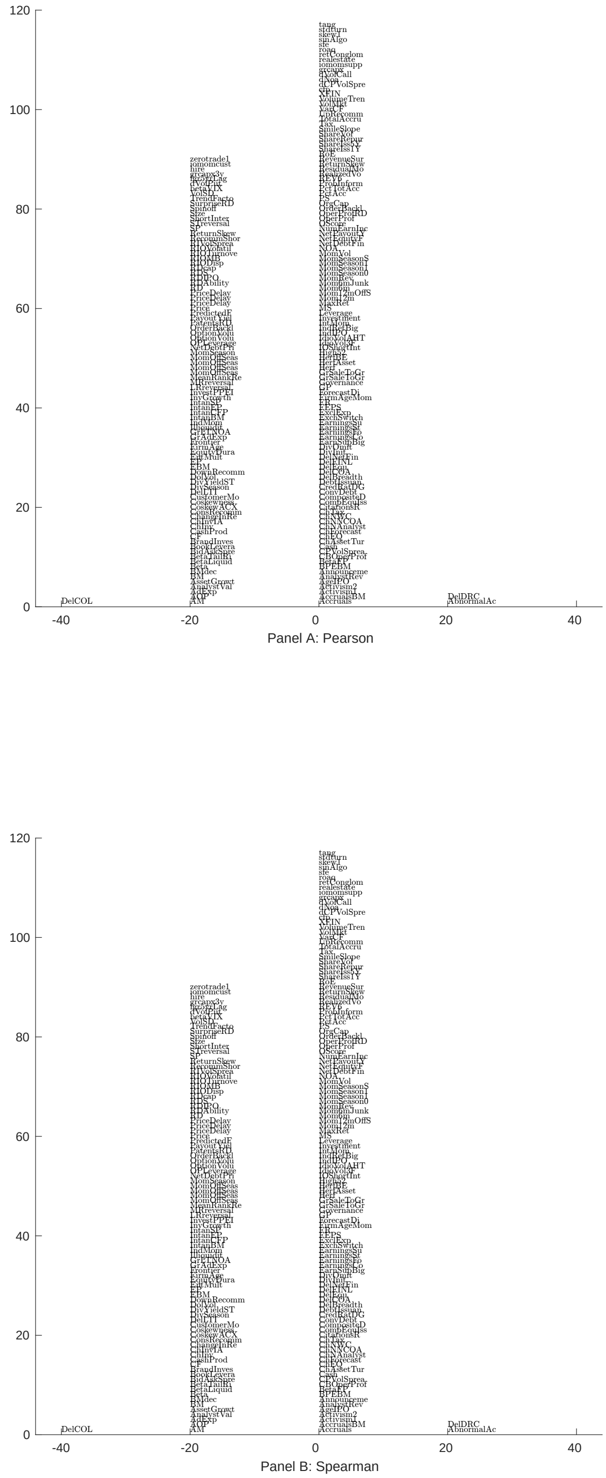


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with ADS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

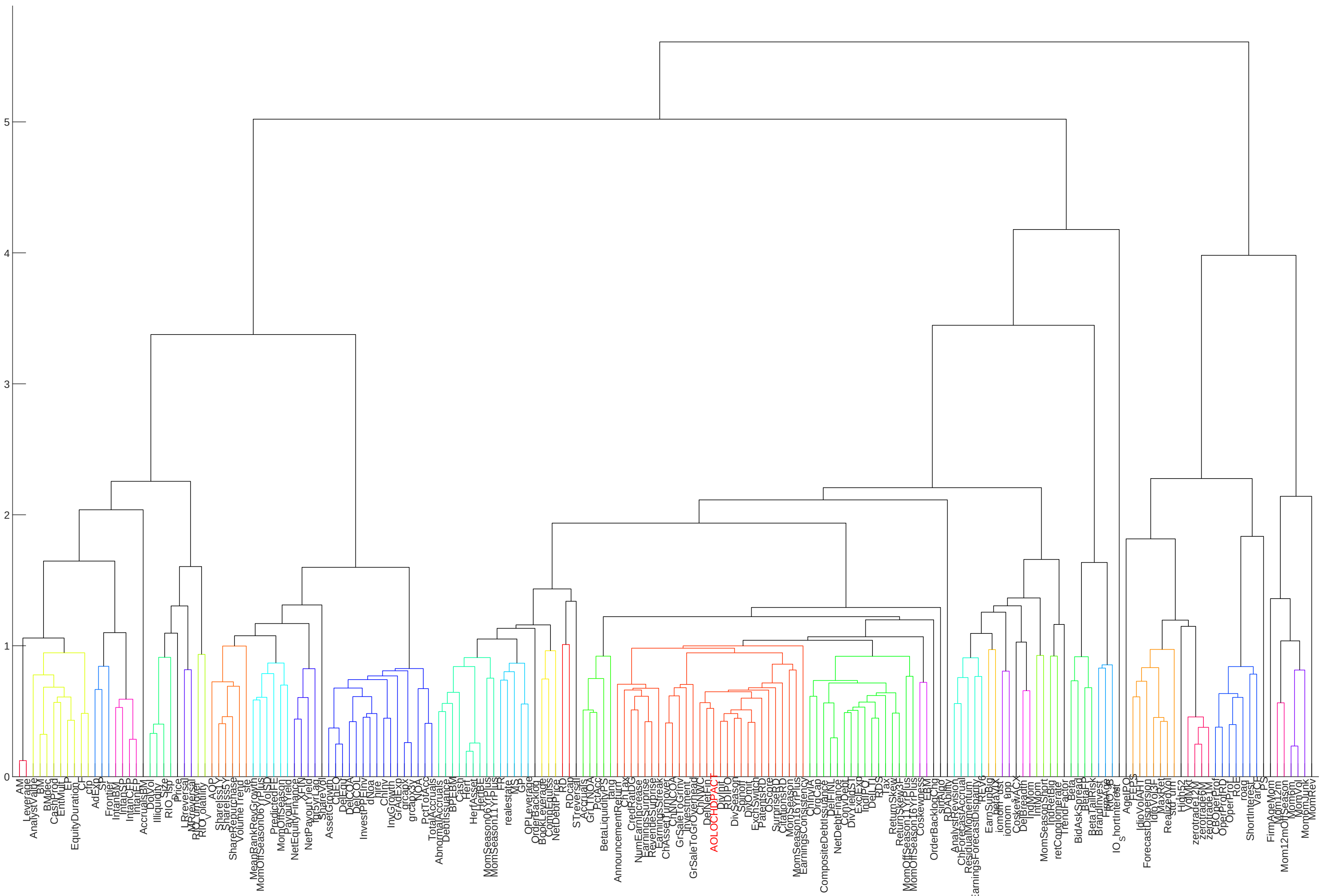


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

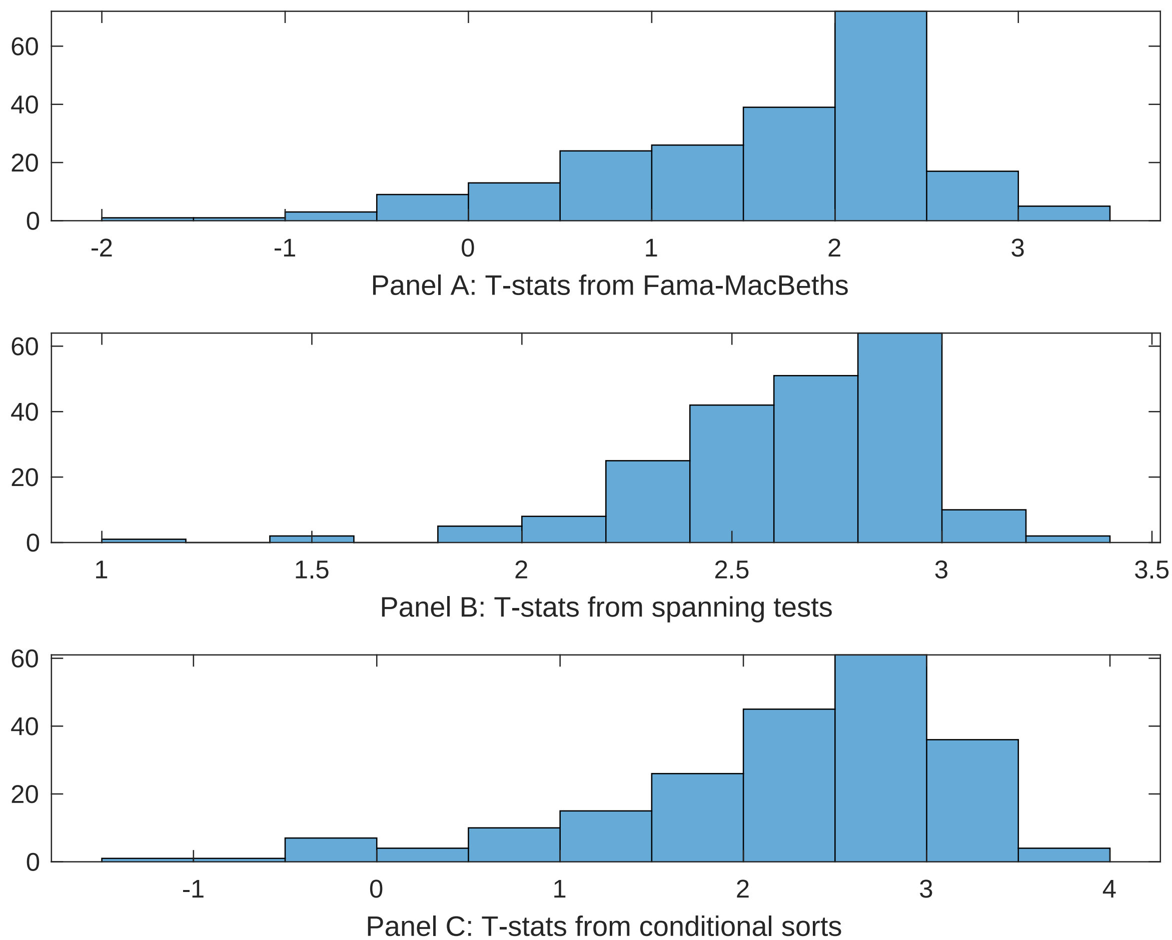


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ADS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ADS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ADS}ADS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ADS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ADS. Stocks are finally grouped into five ADS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ADS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ADS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ADS} ADS_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11	0.99	0.11	0.11	0.12	0.11	0.12
	[3.89]	[3.37]	[3.87]	[3.46]	[3.89]	[3.09]	[3.55]
ADS	0.29	0.60	0.37	0.33	0.57	0.37	0.30
	[1.85]	[2.14]	[2.32]	[2.25]	[1.95]	[2.62]	[0.60]
Anomaly 1	0.50						0.79
	[1.85]						[2.76]
Anomaly 2		0.16					-0.74
		[0.24]					[-0.79]
Anomaly 3			0.51				-0.12
			[0.18]				[-0.27]
Anomaly 4				0.42			-0.21
				[0.26]			[-0.20]
Anomaly 5					0.47		0.37
					[4.84]		[4.33]
Anomaly 6						0.62	-0.29
						[1.17]	[-0.65]
# months	432	427	432	432	427	432	415
$\bar{R}^2$(%)	0	1	0	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the ADS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ADS} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.27	0.30	0.19	0.26	0.29	0.26	0.23
	[2.89]	[3.12]	[2.01]	[2.73]	[3.07]	[2.66]	[2.51]
Anomaly 1	-19.87						-16.13
	[-4.53]						[-3.70]
Anomaly 2		-5.26					-4.55
		[-1.81]					[-1.60]
Anomaly 3			21.17				19.96
			[4.79]				[4.53]
Anomaly 4				-3.41			-11.20
				[-0.64]			[-1.68]
Anomaly 5					-0.63		2.99
					[-0.22]		[0.96]
Anomaly 6						3.28	5.38
						[0.88]	[1.12]
mkt	-2.43	-2.25	-5.10	-3.06	-2.55	-3.31	-6.04
	[-1.03]	[-0.95]	[-2.13]	[-1.25]	[-1.07]	[-1.34]	[-2.31]
smb	-14.32	-10.71	-14.20	-9.01	-12.70	-15.83	-7.30
	[-4.21]	[-2.95]	[-4.19]	[-1.31]	[-3.45]	[-3.26]	[-1.10]
hml	-2.50	1.85	2.17	-0.23	0.03	-0.94	2.02
	[-0.60]	[0.43]	[0.52]	[-0.05]	[0.01]	[-0.22]	[0.47]
rmw	-0.91	5.70	6.09	2.50	4.27	5.39	5.64
	[-0.21]	[1.31]	[1.42]	[0.55]	[0.99]	[1.08]	[1.13]
cma	-4.58	-17.66	-13.55	-18.30	-19.77	-18.81	-3.64
	[-0.68]	[-2.86]	[-2.23]	[-2.97]	[-3.24]	[-3.04]	[-0.54]
umd	4.02	3.72	4.78	4.92	5.31	7.87	4.62
	[1.89]	[1.57]	[2.28]	[1.98]	[1.86]	[2.42]	[1.36]
# months	432	428	432	432	428	432	428
$\bar{R}^2(\%)$	14	11	14	10	11	10	18

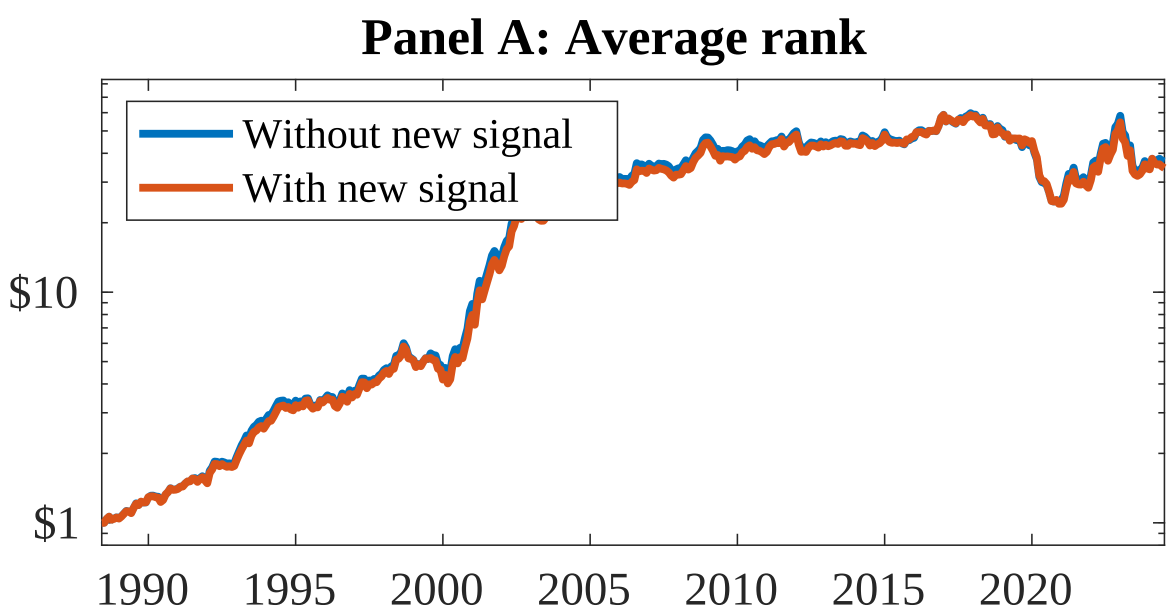


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ADS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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